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Consumer Price Search and Platform Design in Internet Commerce

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1 Big picture

- **Question.** How does **platform design** shape market outcomes in online commerce when consumers face search frictions and sellers strategically set prices?
- **Setting.** The paper studies an important redesign of eBay’s search interface. Before the redesign, buyers saw a **Best Match** results page ranked mainly by relevance. After the redesign, buyers were pushed toward a **product page** that first identified the product and then ranked sellers of that product largely by **price**. The paper combines descriptive evidence, an A/B experiment, and a structural model of search and pricing.
- **Core challenge.** A platform has two jobs that do not always align:
 1. guide consumers to the product they most want, and
 2. intensify price competition among sellers.

Pushing harder on price may reduce markups, but it can also misdirect buyers toward lower-priced, lower-quality offers when products are not truly identical.

- **Main methodological contribution.** The paper builds a model in which the platform affects the **consideration set** consumers actually see, and sellers then set prices in Nash equilibrium taking that search process as given. The key innovation is that browsing data allow the authors to observe consideration sets directly rather than treat them as latent.
- **Why it matters.** In large online markets, the platform is not just a passive intermediary. By changing what consumers see, it changes both demand and seller incentives. That means platform design can affect prices, purchases, markups, and welfare.
- **Main empirical question.** Did eBay’s redesign lower prices because it helped buyers compare sellers more effectively, because it changed what products were bought, or both? And how should the answer depend on product heterogeneity?
- **Main empirical results.**
 - After the redesign, transaction prices fell by about **5–15%** for many product categories.
 - For the structurally estimated product (**Halo Reach**), demand became much more price elastic after the redesign.
 - Median seller margins fell by about **20%**: from \$2.35 to \$1.92 for top-rated sellers and from \$3.48 to \$2.61 for non-top-rated sellers.
 - The single most important force behind stronger price competition was that **price became an important determinant of whether a listing entered the consumer’s consideration set**.
 - The 2012 experiment showed strong **heterogeneity across products**: more heterogeneous products often benefit from relevance-based sorting, while more homogeneous products benefit more from price-based comparison.
 - In counterfactuals, a price-ranked design works well for relatively homogeneous products, but for products with much more vertical differentiation, ranking by quality or demand can outperform ranking by price.

- **Economic takeaway.** Online platform design is a first-order competitive instrument. Reducing search frictions is not just about helping consumers “find something”; it is also about changing how much market power sellers can sustain.

2 Data and measurement (key objects)

2.1 Main data sources

- **Core data.** The paper uses detailed **eBay browsing and transaction data**. A major advantage is that the data show what users actually saw on the platform, making the consideration set observable rather than latent.
- **Broad descriptive analysis.** The redesign is studied in five large product categories: **cell phones, digital cameras, textbooks, video games, and video game systems**. The paper also reports a narrower **iPhone 4** category.
- **Structural estimation sample.** The equilibrium model is estimated on a single, relatively homogeneous product: the Xbox 360 video game **Halo Reach**.
- **Experiment data.** eBay later ran a randomized experiment in 2012 that assigned users either the product-page default or the Best-Match default, allowing a cleaner comparison of search designs on the demand side.

2.2 Institutional design change on eBay

- **Before redesign.** Search queries generated a results page ordered by **Best Match**, a relevance ranking that did not explicitly use price during the period studied.
- **After redesign.** Starting in May 2011, eBay introduced a two-stage search process. Users first chose a product model and then saw a **product page** with listings for that product ordered mostly by **price plus shipping**. A **Buy Box** highlighted the lowest posted price among top-rated sellers.
- **Timing used in the paper.**
 - The product page was introduced on **May 19, 2011**.
 - It became the default for five large categories between **June 27 and July 2, 2011**.
 - The descriptive and structural “after” period begins on **August 1, 2011**, allowing time for seller pricing responses to occur.

2.3 Products, listings, and seller types

- **Targeted vs. non-targeted listings.** For a given search, the platform may show the exact product the buyer wants (targeted listings) and also other related products (non-targeted listings).
- **Top-rated seller (TRS).** eBay gives some sellers a top-rated designation based on sales volume and buyer feedback. This becomes important after the redesign because the **Buy Box** favors top-rated sellers.

- **Halo Reach sample construction.** The estimation focuses on **new, fixed-price Halo Reach listings**. Even after restricting to catalogued Halo Reach listings, some listings are misclassified accessories or non-game items, which is part of why platform filtering matters.

2.4 Key descriptive objects

- **Across broad product categories.**
 - The average number of active listings per product ranges from about **16 to 41**.
 - Price dispersion is large: the ratio of the seventy-fifth to twenty-fifth percentile price is around **1.22** for cell phones, **1.32** for digital cameras, and even larger in other categories.
 - Consumers typically do *not* buy at the lowest posted price; before the redesign, purchased prices often lie around the **25th to 40th percentile** of the price distribution.
- **Halo Reach sample summary.**
 - Before period listings: **270**; after period listings: **218**.
 - Before sellers: **191**; after sellers: **152**.
 - Mean list price falls from **\$39.73** to **\$37.88**; median from **\$37.00** to **\$35.00**.
 - Share of listings from top-rated sellers rises from **16%** to **27%**.
 - Searches rise from **9,427** to **12,059**.
 - New fixed-price Halo Reach purchases rise from **97** to **148**.

2.5 What is actually observed

- **Browsing histories** reveal the listings shown to each buyer on the relevant page.
- **Choices** reveal which listing was purchased, if any.
- **Listing characteristics** include price, seller type, and inferred quality.
- **Prices and sales** allow the authors to back out seller costs from the pricing first-order condition.

3 Models and empirical specifications (all equations + interpretation)

3.1 Conceptual platform-design framework

The paper starts from a general platform model. Let $a_{ij} \in [0, 1]$ denote the probability that consumer i considers product j . The platform controls visibility, subject to attention or display constraints.

Overall demand for product j is

$$D_j(p_j, p_{-j}) = \sum_{L \in 2^{\mathcal{J}}} a_L D_j(p_j, p_{-j}; L),$$

where conditional demand within a consideration set L is

$$D_j(p_j, p_{-j}; L) = \int 1(u(x_j, p_j; \zeta_i) \geq u(x_k, p_k; \zeta_i) \forall k \in L) dF(\zeta_i),$$

and the probability of a particular consideration set is

$$a_L = \left(\prod_{j \in L} a_j \right) \left(\prod_{j \notin L} (1 - a_j) \right).$$

Interpretation. The platform changes demand not by changing utility directly, but by changing *what consumers get to compare*. This is the key object that links platform design to both consumer behavior and seller pricing incentives.

3.2 Seller pricing and the two margins of platform design

Seller j solves

$$\pi_j = \max_{p_j} D_j(p_j, p_{-j})((1-t)p_j - c_j - T),$$

which gives the first-order condition

$$p_j = \frac{c_j + T}{1-t} - \left(\frac{\partial D_j(p_j, p_{-j})}{\partial p_j} \right)^{-1} D_j(p_j, p_{-j}).$$

The derivative of demand with respect to price decomposes as

$$\frac{\partial D_j(p_j, p_{-j})}{\partial p_j} = \sum_L a_L \frac{\partial D_j(p_j, p_{-j}; L)}{\partial p_j} + \sum_L \frac{\partial a_L}{\partial p_j} D_j(p_j, p_{-j}; L).$$

Interpretation. Price affects seller demand in **two** ways:

1. the usual **within-consideration-set** effect: conditional on being seen, a lower price makes purchase more likely;
2. the **platform-selection** effect: if lower price makes a listing more likely to be shown, then lower prices matter even before the consumer chooses.

This second channel is the paper's central mechanism.

3.3 Toy example: why more price ranking is not always better

In the stylized example, there are two products with qualities $q_1 = q > q_2 = 0$ and costs $c_1 = c$, $c_2 = 0$. Consumer utility is

$$u_{ij} = \zeta_i + q_j - \alpha p_j,$$

with outside option normalized to zero.

Demand for product j is

$$D_j(p_j, p_{-j}) = \begin{cases} a_j(p_j, p_{-j}) & \text{if } p_j < q_j/\alpha, \\ a_j(p_j, p_{-j})(1 + q_j - \alpha p_j) & \text{if } p_j \in [q_j/\alpha, (1 + q_j)/\alpha], \\ 0 & \text{if } p_j > (1 + q_j)/\alpha. \end{cases}$$

The platform shows product 1 with probability

$$a_1 = \frac{\exp((q - \beta p_1)/\sigma)}{\exp((q - \beta p_1)/\sigma) + \exp((- \beta p_2)/\sigma)},$$

and $a_2 = 1 - a_1$.

Interpretation. The parameter β is the platform's weight on price in determining visibility:

- if $\beta = 0$, the platform cares only about non-price attributes;
- as β rises, cheaper products are shown more often.

The toy model illustrates the paper's main trade-off:

- higher β makes seller demand more price-sensitive and lowers markups;
- but higher β also overexposes cheaper, lower-quality products, which can distort matching.

Hence the consumer-surplus-maximizing β^* is generally **interior**, not zero and not infinity.

Comparative statics from the toy model:

- β^* rises with consumers' price sensitivity α ;
- β^* rises when the platform is noisier in measuring quality (σ larger);
- β^* falls when vertical differentiation q is larger;
- β^* falls when the high-quality seller's cost c is higher.

Economic meaning. Price ranking is most attractive when products are close substitutes and the platform is poor at reading non-price quality. It is less attractive when product quality differences are important.

3.4 Empirical consumer demand model

For a targeted listing j , utility is specified as

$$u_{ij} = \alpha_0 + \alpha_1 p_j + \alpha_2 \text{TRS}_j + \alpha_3 p_j \text{TRS}_j + \alpha_4 q_j + \varepsilon_{ij},$$

where ε_{ij} is Type I extreme value.

For a non-targeted listing m ,

$$u_{im} = \delta + \lambda \varepsilon_{im}.$$

The outside option is

$$u_{i0} = \varepsilon_{i0}.$$

Interpretation.

- p_j captures how price lowers demand.
- q_j is listing quality, inferred from before-period Best-Match exposure.
- TRS_j captures the demand value of being a top-rated seller.

- λ determines how differentiated non-targeted products are from each other and from the targeted product.

Because the browsing data reveal what the buyer actually saw, demand estimation is a standard multinomial logit conditional on observed consideration sets.

3.5 Consideration set formation

Let $L_i = (L_i^J, L_i^M)$ denote the number of targeted and non-targeted items in consumer i 's consideration set. The distribution of L_i is estimated directly from browsing data.

Before redesign. Listings are sampled according to quality. The idea is that Best Match is approximately a quality-ranking rule rather than a price-ranking rule.

After redesign. Sampling weights become

$$\omega_j = \exp \left[-\gamma \left(\frac{p_j - \min_{k \in \mathcal{J}_i^J}(p_k)}{\text{std}_{k \in \mathcal{J}_i^J}(p_k)} \right) \right].$$

A larger γ means that lower-priced listings are more likely to enter the page the buyer sees. The redesign also reserves one spot in the consideration set for a **TRS** listing through the Buy Box.

Interpretation. The parameter γ measures how aggressively the platform translates low prices into visibility.

3.6 Pricing behavior and cost recovery

Given demand and the search process, seller-specific costs are backed out from the before-period first-order condition as

$$c_j = (1 - t)p_j - T + (1 - t) \frac{D_j(p_j)}{D'_j(p_j)}.$$

Interpretation. The authors do not observe marginal costs directly. Instead, they ask: what cost would make the observed price optimal, given the estimated demand system and consideration-set process?

This delivers an implied distribution of seller costs, including some extremely high inferred costs for sellers posting very high prices.

3.7 Counterfactual platform analysis

With demand and costs estimated from the before period and the consideration-set process estimated for the after period, the paper simulates equilibrium outcomes under alternative search designs:

- quality rank (before-type design),
- price rank (after-type design),
- demand-weight rank,

- no search frictions (all items visible).

It also varies product differentiation to study when price ranking is useful and when it is not.

Interpretation. This is where the paper moves from documenting one redesign to asking a more general policy question: how should an online platform rank products?

4 Identification and instruments (what makes the estimates credible)

4.1 What standard approaches miss

- Standard transaction data do not reveal what the buyer actually saw, so the consideration set must be treated as latent.
- Without browsing data, it is difficult to separate search frictions from tastes, seller heterogeneity, and pricing power.
- A small-scale experiment can identify some short-run demand responses, but not equilibrium price responses if sellers set one price for all users.

4.2 What identifies demand and search frictions

- **Browsing data** identify consumer consideration sets directly.
- The **before period** helps identify listing quality because Best Match did not explicitly use price in ranking during the study period.
- The **after period** identifies γ , the extent to which lower prices raise visibility on the product page.
- The presence of **TRS** sellers and the Buy Box identifies the demand and visibility advantage of being top-rated.

4.3 What identifies supply, costs, and markups

- Given estimated demand and the observed before-period pricing equilibrium, seller costs are backed out from the sellers' first-order conditions.
- This is possible because the paper combines consumer-side information (choices and consideration sets) with seller-side pricing behavior.
- The after-period redesign is then evaluated by changing only the search process, holding demand primitives and cost distributions fixed.

4.4 Why the experiment is not enough on its own

- The 2012 A/B experiment changed defaults for only a subset of users, so sellers did not face different equilibrium pricing incentives across experimental groups.
- Sellers also cannot instantaneously reoptimize in response to a small, short-lived experiment.

- Therefore, the experiment is useful for showing heterogeneity in demand-side effects across products, but not for measuring the full equilibrium effect on prices and seller margins.

4.5 Credibility and modeling choices

- The authors deliberately focus structural estimation on a relatively homogeneous product (**Halo Reach**) so that price comparison is meaningful and product classification is tractable.
- They use before-period data for demand estimation and then ask whether the changed search design can predict after-period prices and quantities. This is a strong out-of-sample check.
- The model predicts several moments well, including the Buy Box purchase share and the after-period decline in prices.

4.6 Relation to the literature

- It connects to the empirical search literature through observable consideration sets.
- It contributes to the industrial-organization literature on price dispersion and markups in online markets.
- It also contributes to the two-sided-platform literature by emphasizing that a platform’s ranking rule shapes seller competition, not just buyer navigation.

5 Tables and figures

5.1 Figure 1 and Figure 2: the toy trade-off and optimal platform design

Read:

- Figure 1 shows that increasing the platform’s weight on price lowers seller prices, but also makes the cheaper and lower-quality product more likely to be shown.
- Consumer surplus is therefore hump-shaped in the price weight β .
- Figure 2 shows the comparative statics of the optimal β^* .

Punchline. The optimal platform design is neither “always show the highest-quality product” nor “always show the cheapest product.” It depends on demand elasticity, cost differences, vertical differentiation, and the noisiness of the platform’s quality signal.

5.2 Figure 3 and Figure 4: alternative platform designs and the eBay redesign

Read:

- Figure 3 compares three search philosophies:
 - Craigslist: broad, inclusive, weakly filtered list;
 - Amazon: product-level classification first, then seller comparison;

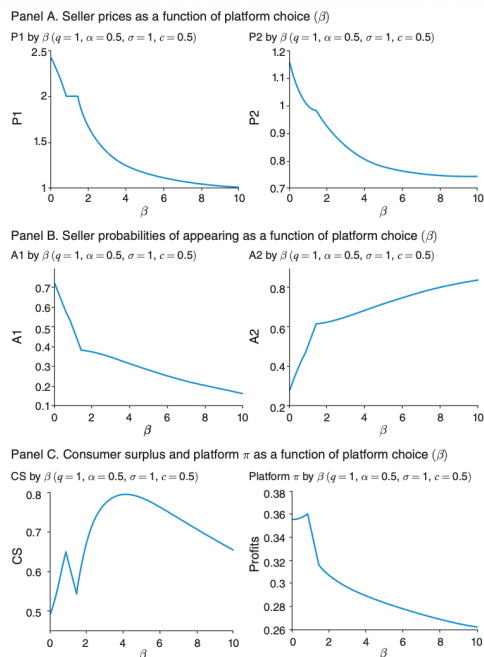


FIGURE 1. COMPARATIVE STATICS IN PLATFORM CHOICE

Notes: Figure shows seller prices (panel A), seller ex ante probabilities of being shown (panel B), and consumer surplus and profits (panel C) as a function of the platform's choice of the relative weight (β) on price when determining which product to show users.

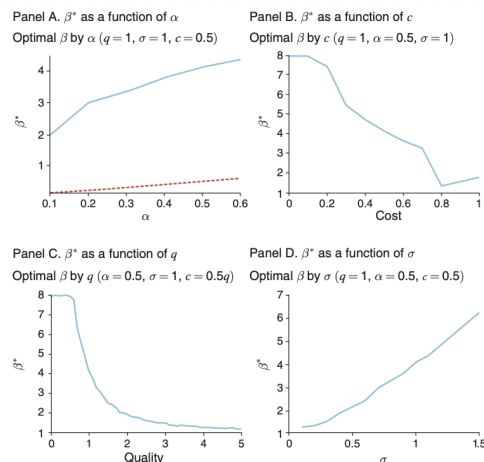


FIGURE 2. (CONSUMER SURPLUS) OPTIMAL PLATFORM CHOICE

Notes: Figure shows β^* , the platform's choice of the relative weight to put on price that maximizes consumer surplus, as a function of consumers' price sensitivity in demand (panel A), the cost of the higher-quality product (panel B), the quality difference between the products (panel C), and the platform's noise in observing quality (panel D). Parameters are fixed at $\alpha = 0.5, q = 1, c = 0.5q$, and $\sigma = 1$ unless they are the parameter being varied for the comparative static. The platform charges 0.3 per transaction and keeps 0.1 of the transacted price.

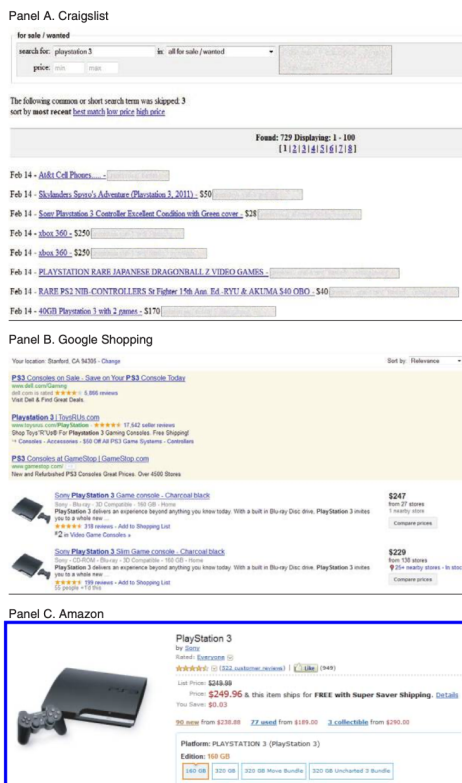


FIGURE 3. DIFFERENT APPROACHES TO PLATFORM DESIGN

Note: Figure shows search results following a query for "Playstation 3" on Craigslist (panel A), Google Shopping (panel B), and Amazon (panel C).

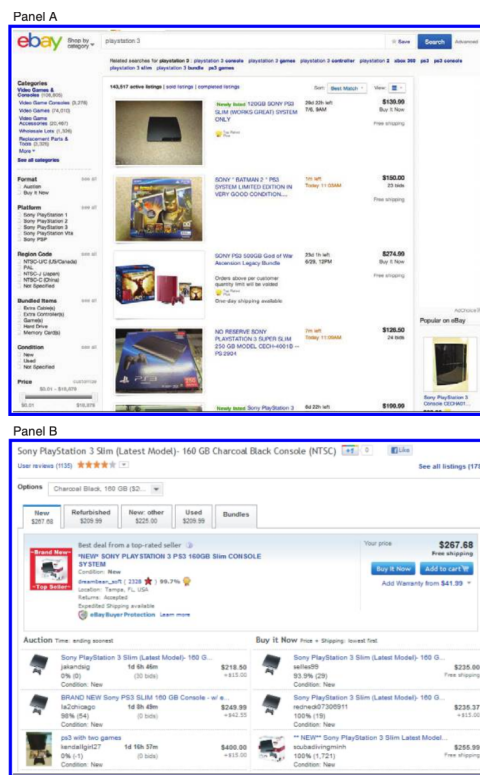


FIGURE 4. EBAY'S PLATFORM REDESIGN

Notes: Figure shows the change in eBay's presentation of search results. The top panel shows eBay's Best Match results. The bottom panel shows a product page, with listings ordered by sales format and price.

– Google Shopping: an intermediate case.

- Figure 4 shows eBay’s move from the Best-Match results page to the product page with a Buy Box and price-ranked listings.

Punchline. The paper is about a concrete institutional shift from relevance-style ranking toward price-based seller comparison within a product page.

5.3 Table 1 and Figure 5: descriptive evidence on the redesign

TABLE 1—CATEGORY-LEVEL EFFECTS OF THE PLATFORM REDESIGN							
		Cell Phones	Digital Cameras	Textbooks	Video Game Systems	Video Games	iPhone 4
Average number of active listings	Before	23.33	40.64	16.27	35.41	38.93	29.49
75th/25th percentile of Posted prices	Before	1.22	1.32	1.61	1.39	1.47	1.28
Transacted prices	Before	204.10	1.31	1.25	1.29	1.29	138.44
Average price percentile of bought items	Before	40.11	31.85	29.82	17.34	27.82	37.40
	After	37.79	24.94	20.75	19.72	19.22	33.07
	Change	−2.32	−6.90	−9.07	2.38	−8.61	−4.33
		(0.98)	(1.21)	(1.76)	(0.84)	(0.61)	(0.62)
Posted prices (\$, mean)	Before	562.45	1,418.31	67.98	290.00	48.60	749.25
	After	462.88	1,170.81	63.11	285.08	48.14	567.74
	Change	−99.57	−247.50	−4.86	−4.92	−0.45	−181.50
		(11.47)	(13.14)	(2.61)	(5.32)	(0.84)	(12.04)
Transacted prices (\$, mean)	Before	412.30	1,162.51	51.03	222.62	45.71	676.72
	After	403.75	980.06	42.89	257.16	42.17	554.82
	Change	−8.55	−182.46	−8.14	34.53	−3.53	−121.90
		(11.26)	(10.78)	(0.59)	(3.47)	(0.40)	(6.83)
Number of transactions	Before	1,762	650	482	1,045	3,873	2,605
	After	3,594	3,108	3,941	1,666	2,537	4,346
	Change	1,832	2,458	3,459	621	−1,336	1,741
TRS share of transactions (%)	Before	43.87	70.92	27.39	43.92	27.11	36.62
	After	39.12	78.93	45.27	42.62	44.42	40.52
	Change	−4.75	8.00	17.88	−1.31	17.31	3.90
		(1.44)	(1.93)	(2.18)	(1.96)	(1.22)	(1.20)

Notes: Table presents statistics at the category level before and after the product page introduction. The Before period spans April 6, 2011 to May 18, 2011; the After period spans August 1, 2011 to September 20, 2011. For each category we choose the ten products that appeared most often in search results during the week before July 2, and report statistics based on a weighted average across these ten products. To calculate the price percentiles of bought items: for each purchase, we find all the listings that were available at the time of purchase, and use the percentile in this distribution. We note that we use eBay’s classification of listings to product identifiers. In some cases (most commonly in the context of cell phones), accessories get classified as the product itself, leading to what may appear as large price dispersion, which more likely reflects product misclassification.

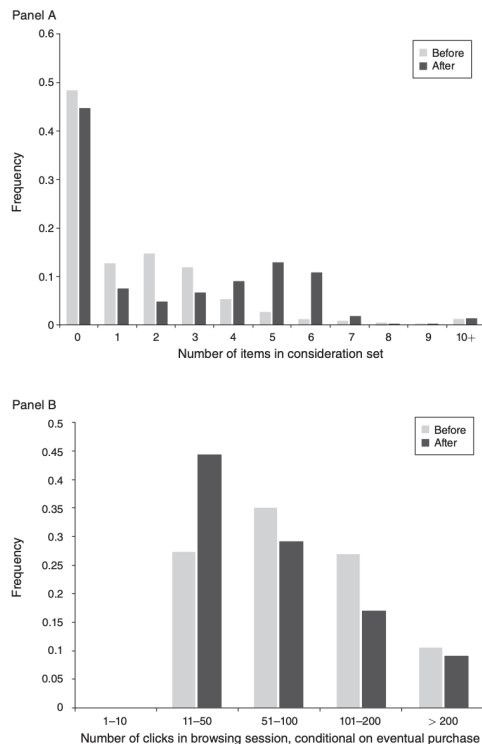


FIGURE 5. CHANGE IN SIZE OF CONSIDERATION SET

Notes: Figure shows changes in browsing experience between the Before (April 6th to June 18th) and After (August 1st to September 20th) periods of 2011. Top panel shows distributions of the size of the targeted consideration set, L^* ; that is, the number of targeted items shown on the search results page (the default in the Before period) or the product page (the default in the After period) for *Halo Reach* listings. For users that visited multiple pages, the consideration set includes all listings on the pages. Bottom panel plots the distribution of clicks per search session prior to eventual purchase of a targeted (i.e., new, fixed price) *Halo Reach* listing. A click counts if it led to eBay loading a page, and counting starts from the first “Halo Reach” search event.

Read:

- Across five broad categories, transaction prices fall after the redesign in every category except video game systems.
- The fall is small in some categories (about **2.1%** for cell phones and **7.7%** for video games) and large in others (about **15.7%** for digital cameras and **15.9%** for textbooks).
- Posted prices also fall in every category except video game systems, by between roughly **0.9%** and **17.7%**.
- Buyers shift toward cheaper offers and toward top-rated sellers.

- Figure 5 shows that the Halo Reach consideration set becomes much larger after the redesign, and eventual buyers need fewer clicks before purchase.

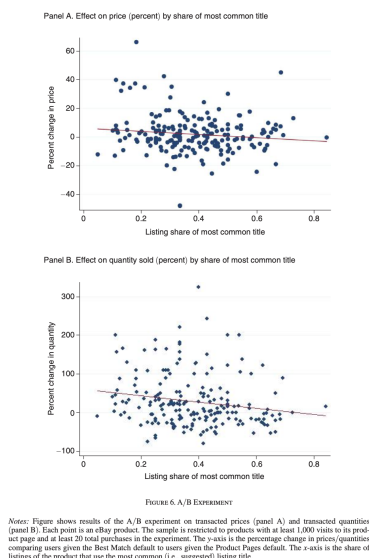
Punchline. The redesign changes both **what consumers buy** and **what sellers charge**. Search became more streamlined, and seller pricing became more competitive.

5.4 Figure 6: the 2012 A/B experiment

Read:

- When users are randomized between Best Match and product-page defaults, the average effects on prices and quantities are small.
- But there is substantial heterogeneity across products.
- Products with more heterogeneous listings benefit more from Best Match, while products with more standardized listings show little quantity gain and even a slight negative price effect under Best Match.

Punchline. This directly supports the paper's central idea that the best ranking rule depends on product differentiation.



	Before	After
<i>Panel A. Listings data</i>		
Number of listings	270	218
Number of sellers	191	152
Percent of sellers with >1 listing	20	22
Mean list price (+ shipping)	\$59.73	\$37.88
Median list price (+ shipping)	\$37.00	\$35.00
Standard deviation of list price (+ shipping)	\$9.20	\$8.73
Percent of listings from TRS	16	27
Mean quality	0.037	
Median quality	0.002	
<i>Panel B. Search data</i>		
Number of "search results page" searches	9,409	5,344
Number of product page searches	18	6,715
Total number of searches	9,427	12,059
Mean transacted price (+ shipping)	\$34.56	\$33.30
Median transacted price (+ shipping)	\$34.99	\$34.00
Standard deviation of transacted price (+ shipping)	\$2.59	\$3.89
Mean quality	0.111	
Median quality	0.021	
Number of Halo Reach fixed price new transactions	97	148
Number of other transactions	185	317

Notes: Before: April 6, 2011 to May 18, 2011. After: August 1, 2011 to September 20, 2011. Panel A uses listing-level data. Panel B uses search-level data. Targeted listings are considered to be the correct product if they are listed with the Halo Reach product code and inspection of their title indicates that the listing is not for an accessory. "TRS" refers to top-rated sellers, an eBay designation that depends on a seller's volume and feedback.

	Before	Predicted after
<i>Platform Parameters</i>		
Average number of listings on the site	21	28
Average number of TRS listings on the site	3	10
Prob. of a single-targeted item consideration set	0.13	0.08
Estimated gamma		0.80
		(0.14)
Median prob. of appearing in a search		0.08
Prob. of appearing if lower price by 10%		0.11
<i>Demand</i>		
Constant (Halo Reach new fixed price)	3.72	
	(1.07)	
Price	-0.24	
	(0.03)	
Top-rated seller (TRS)	4.13	
	(2.75)	
Price $\times$ TRS	-0.10	
	(0.08)	
Quality (0 to 1)	0.64	
	(0.36)	
Constant (Other listings)	-8.37	
	(0.41)	
Size of episode (Other listings)	1.70	
	(0.14)	
<i>Implied Price Elasticities</i>		
Average own-price elasticity	-10.64	-13.53
Average own-price TRS elasticity (TRS)	-14.16	-17.09
Average own-price TRS elasticity (Non-TRS)	-9.95	-12.84
<i>Supply</i>		
Median price - cost (TRS)	\$2.35	\$1.92
Median margin (percent of P) (TRS)	0.06	0.05
Median price - cost (Non-TRS)	\$3.48	\$2.61
Median margin (percent of P) (Non-TRS)	0.09	0.07
<i>Purchase Rates (percent)</i>		
(Halo Reach) observed	1.03	1.23
(Halo Reach) predicted	0.93	1.47
(Other) observed	1.96	2.63
(Other) predicted	1.27	1.85
(Buy Box, conditional on HR purchase) observed		64
(Buy Box, conditional on HR purchase) predicted		65

Notes: Estimates of demand model parameters use data from the Before period only (estimated standard errors in parentheses). The remaining statistics are calculated from these estimates. The implied price elasticities and pricing predictions for the After period use browsing data from the After period as described in the main text. The Halo Reach and Other purchase rates are defined as the shares of relevant search queries that end up transacting in a targeted (new, fixed price) Halo Reach listing or other listing, respectively. The Buy Box purchase rates are the percentage of Halo Reach purchases that came from the Buy Box listing.

5.5 Table 2 and Table 3: Halo Reach sample and structural estimates

Read:

- Halo Reach is a relatively homogeneous product with many competing sellers and substantial price variation.
- The after period has more TRS listings and lower prices.
- The estimated search parameter is $\gamma = 0.80$, so a **10%** price cut makes a listing about **27%** more likely to enter the consideration set.

- Demand is very price elastic: average own-price elasticity is -10.64 before and -13.53 after; for TRS sellers the elasticity rises from -14.16 to -17.09 .
- Median seller margins are economically meaningful but not huge: **6–9%** of price before the redesign.
- The model predicts the Buy Box share of Halo Reach purchases almost exactly: **65% predicted versus 64% observed**.

Punchline. The structural model fits the data reasonably well and implies that search frictions and ranking rules matter a lot for pricing power even in a relatively homogeneous product market.

5.6 Table 4 and Figures 7–9: what part of the redesign mattered?

	Supply: Median price cost margins				HR purchase rate predicted (%)	Platform profit (\$)
	TRS (\$)	Non-TRS (\$)	TRS (%)	Non-TRS (%)		
<i>Panel A. Implementing the platform change</i>						
Before	2.35	3.48	6	9	0.9	0.08
Larger consideration set	2.35	3.48	6	9	1.0	0.12
Additional sellers	2.35	3.48	6	9	1.1	0.08
Buy Box	2.35	3.48	6	9	1.0	0.08
Increase in gamma	1.90	2.66	5	7	1.1	0.08
Predicted after	1.92	2.61	5	7	1.5	0.12
<i>Panel B. Implementing the platform change without price changes</i>						
Before prices, after design	2.35	3.48	6	9	1.3	0.12

Notes: The top and bottom rows of the panel A report the margins and purchase rate of targeted listings from the estimated model, as shown in Table 3. The middle rows break down the effect of the platform change by starting from the before parameters and separately increasing consideration sets, adding additional sellers, introducing a Buy Box that samples a TRS listing, and increasing price-dependence in the search. Panel B estimates the effect of the platform change on purchase rates while keeping the prices at the before levels.

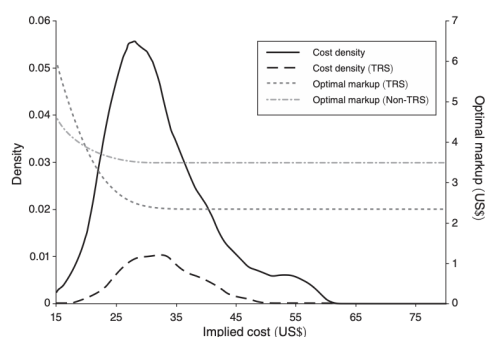


FIGURE 7. IMPLIED COST DISTRIBUTION

Notes: Figure shows the distribution of seller costs imputed from the observed prices and the sellers' first-order condition. This cost distribution is assumed to remain the same after the platform redesign, and is held fixed in the counterfactual exercises. The dashed black line shows the cost distribution of TRS sellers; the solid black line is non-TRS sellers. The optimal mark-ups associated with each level of cost, given our demand estimates, are presented by the gray lines for TRS and non-TRS.

Read:

- Figure 7 shows a wide implied cost distribution across sellers.
- Figure 8 shows demand curves becoming notably more elastic after the redesign, especially for TRS sellers.
- Figure 9 shows the model reproduces the shift in the observed price distribution after the redesign.
- Table 4 decomposes the redesign. The critical result is that making price matter more in visibility (raising γ) is the dominant force lowering margins.
- The median TRS margin falls from **\$2.35** to **\$1.92**; the median non-TRS margin falls from **\$3.48** to **\$2.61**.
- About **62%** of the increase in purchase rates comes from the redesign even if prices are held fixed, while the remaining **38%** comes from sellers' price response.

Punchline. The redesign worked mainly because it altered the mapping from prices to visibility, thereby changing seller incentives. Search design affected equilibrium, not just clicking behavior.

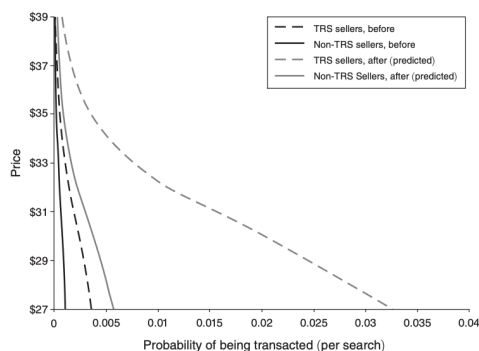


FIGURE 8. ESTIMATED DEMAND CURVES

Notes: Figure plots demand curves based on our model estimates. The x-axis is the per-search probability of being transacted, which is the probability of appearing in the consideration set multiplied by the probability of being transacted conditional on being in the consideration set.

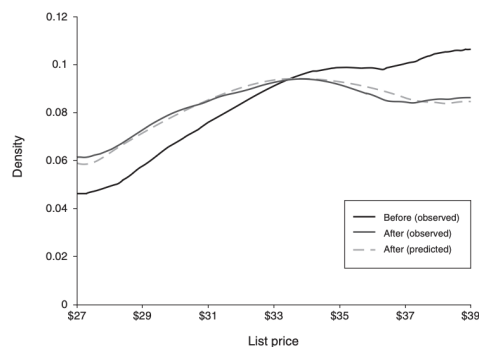


FIGURE 9. OBSERVED AND PREDICTED PRICE DISTRIBUTIONS

Notes: Figure shows distributions of posted prices from the Before and After periods, and the predicted price distribution for the After period based on our estimated model. Note that the model is estimated using only Before data (except for the use of the After data to estimate the size of the consideration set and parameter γ).

5.7 Table 5 and Table 6: broader counterfactuals and high-cost sellers

Read:

- Table 5 compares alternative search designs under different degrees of product differentiation.
- For relatively homogeneous products with limited seller differentiation, price-rank search lowers markups and prices relative to quality-rank search.
- With no search frictions and limited differentiation, median markups fall below **\$0.50**, showing how strongly search frictions contribute to pricing power.
- When horizontal differentiation is important, a price-rank design can generate *more* intense competition than a no-frictions design because the limited consideration set itself is selected using price.
- For products with much greater vertical differentiation, however, **quality rank or demand-weight rank** can outperform price rank.
- Table 6 explores the high inferred costs of some Halo Reach sellers and finds only weak partial explanations.

Punchline. There is no universally optimal ranking rule. Price ranking is especially valuable for standardized products, but it can become too aggressive when products differ meaningfully in quality.

6 Short summary

- **Method.** Build a structural model of consumer search and seller pricing in which the platform changes the consumer's consideration set.
- **Key data advantage.** Browsing data make consideration sets observable.
- **Application.** eBay's redesign from Best Match to product-page search.
- **Main result.** The redesign reduced prices mainly because it made low prices more likely to be shown to buyers, which sharply increased sellers' effective demand elasticities.

TABLE 5—THE IMPACT OF SEARCH FRICTIONS

		Search Design			No frictions (all items visible)
		Quality rank (Before)	Price rank (After)	Demand weights rank	
<i>Panel A. Using estimated quality</i>					
Limited seller differentiation	Median markup	\$1.95	\$1.71	\$1.71	\$0.45
	Mean transacted price	\$33.37	\$32.03	\$32.04	\$24.61
	Purchase rate (HR)	1.1%	1.4%	1.4%	2.6%
	Purchase rate (other)	2.4%	2.2%	2.2%	2.1%
	Platform profit	\$0.12	\$0.12	\$0.12	\$0.14
Seller differentiation	Median markup	\$3.57	\$2.58	\$2.59	\$3.46
	Mean transacted price	\$34.70	\$33.02	\$33.05	\$33.56
	Purchase rate (HR)	1.2%	1.4%	1.4%	2.2%
	Purchase rate (other)	2.1%	2.0%	2.0%	1.7%
	Platform profit	\$0.12	\$0.12	\$0.12	\$0.12
<i>Panel B. Using quality with 30 times the dispersion of estimated quality</i>					
Limited seller differentiation	Median markup	\$2.90	\$2.24	\$2.85	\$0.90
	Mean transacted price	\$37.84	\$36.33	\$37.79	\$40.61
	Purchase rate (HR)	2.7%	2.6%	2.7%	9.8%
	Purchase rate (other)	2.3%	2.5%	2.2%	1.6%
	Platform profit	\$0.19	\$0.19	\$0.19	\$0.48
Seller differentiation	Median markup	\$3.46	\$2.59	\$3.45	\$3.46
	Mean transacted price	\$38.24	\$36.55	\$38.20	\$40.26
	Purchase rate (HR)	2.7%	2.6%	2.8%	7.0%
	Purchase rate (other)	2.2%	2.4%	2.1%	1.6%
	Platform profit	\$0.19	\$0.18	\$0.19	\$0.36

Notes: Panel A uses the estimated quality distribution from the *Halo Reach* product used in estimation. Panel B considers a hypothetical product with 30 times the dispersion in quality of *Halo Reach*. Seller differentiation and limited seller differentiation refer to whether we include a seller-specific logit error for targeted listings. The version with differentiation keeps the error, while the limited differentiation specification assumes a nested logit model in which all “targeted listings” are in the same nest and the nested logit σ parameter is set to 0.8. Each column refers to a different platform design: quality rank (the Before regime), price rank (the After regime), a counterfactual regime with demand weights rank, and a counterfactual regime in which consumers are shown the entire set of targeted listings available on the platform.

TABLE 6—EXPLANATION OF HIGH SELLER COSTS

	Univariate regressions	
	Dummy for seller's max cost > \$40	
Top-rated seller dummy	−0.024	(0.080)
Years on eBay (truncated at 5)	0.006	(0.036)
Listings fixed price (percent)	−0.176	(0.070)
Listings free shipping (percent)	−0.152	(0.071)
log (total <i>Halo Reach</i> listings)	0.040	(0.020)
log (total video game listings)	0.008	(0.009)
log (total eBay listings)	0.010	(0.009)
log (Q available in all listings)	0.012	(0.008)
log (Q sold in all listings)	0.000	(0.010)
log (number categories listing in)	0.002	(0.022)
Any <i>Halo Reach</i> price change	0.080	(0.070)

Notes: Table shows results from univariate regressions where each observation is a seller in the before period ($N = 191$) and the dependent variable is an indicator equal to 1 if the seller's imputed cost from the model is above \$40 for at least one of his *Halo Reach* listings. The covariates pertaining to characteristics of seller listings are generated using all listings by the seller over the period 2009–2011.

- **Broader lesson.** Platform design should depend on product heterogeneity: price-based ranking is most useful for relatively homogeneous goods.

7 Conclusion

7.1 Core conclusion (what the paper establishes)

The paper shows that online platform design is a central determinant of market outcomes. By changing which offers consumers actually see, a platform changes both consumer choices and the competitive pressure sellers face. In the eBay application, moving toward a product page that emphasized low prices lowered seller margins and transaction prices, but the desirability of such a design depends strongly on product heterogeneity.

7.2 Economic intuition

A platform ranking rule changes seller competition through an extra margin beyond standard demand substitution. If a low price not only makes a seller more attractive conditional on being seen, but also makes the seller *more likely to be seen at all*, then price competition becomes much tougher. This is why price-based ranking can have effects that are larger than what standard demand models alone would imply.

At the same time, price-based ranking is not always efficient. When products differ materially in quality, features, or fit, ranking purely by price can send consumers toward cheaper but worse matches. The right platform design balances **matching efficiency** against **competitive pressure**.

7.3 Takeaway

This paper turns platform design into an equilibrium object in industrial organization. The main lesson is simple but powerful: **what a platform chooses to show buyers is also a policy choice about how hard sellers compete**. In standardized markets, making price prominent can substantially reduce markups. In differentiated markets, relevance and quality still matter.